Define data compression and writes its necessity. Describe different coding schemes.Write the steps of Huffman coding.

Data Compression
Coding Schemes
Huffman Coding Steps

Definition of Data Compression

Image compression reduces file size by removing redundant data, enabling efficient storage and transmission.

Necessity of Compression:

- Data storage
- Data transmission

The digital information revolution has brought an explosive increase in the amount of data.Digital data requires large storage and high transmission bandwidth.

Compression is essential to facilitate storage capacity and economical rapid transmission over low bandwidth lines

Coding Schemes

Huffman Coding: Optimal lossless method using a probability-based tree to assign shorter codes to frequent symbols.

LZW: Dictionary-based coding that builds a sequence library dynamically without needing prior probability data.

Run-Length Coding : Replaces identical consecutive symbols with a pair, best for repetitive data.

Symbol-Based Coding: Stores recurring sub-images in a dictionary and encodes their locations in the image.

Bit-Plane Coding: Splits images into binary layers and compresses each separately; highly sensitive to intensity changes.

Block Transform Coding: Divides images into blocks for mathematical transformation and quantization; used in JPEG.

Huffman Coding Steps:

1.List all symbols with their probabilities in descending order.
2.Combine the two symbols with the lowest probabilities into a new node.
3.Re-sort the list.
4.Repeat steps 2–3 until only two symbols/nodes remain.
5.Assign binary codes by traversing the tree: 0 to one branch, 1 to the other.
6.The code for each symbol is read from root to leaf.